

## Problem Permuturns

Input file      `stdin`  
Output file    `stdout`



Figure 1: The tarot reading of the problem *Permuturns*

Alin and Blin are playing a game called Palatro. They share a deck of  $N$  cards labeled 1 through  $N$ . Alin writes the number  $A_i$  on card  $i$ , and Blin writes the number  $B_i$  on card  $i$ . Each player must write every number from 1 to  $N$  on exactly one card (thus  $A$  and  $B$  are permutations).

Clin is the referee of the game. He decides the order in which the two players must reveal their cards, giving  $N$  requests of the form:

“Show the card with index  $C_1$ ”, then “Show the card with index  $C_2$ ”, and so on. Here  $(C_1, C_2, \dots, C_N)$  represents a permutation of the numbers from 1 to  $N$ .

At the beginning, Clin considers the first card shown ( $C_1$ ) to be the winning card. Each time a new card  $C_i$  is revealed, Clin checks the numbers written on it: if both Alin’s number and Blin’s number are greater than those on the current winning card, then card  $C_i$  becomes the new winning card.

More formally, Clin uses the following algorithm to determine the winning card:

```
winner = C[1]
for i=2...N:
    if A[C[i]] > A[winner] and B[C[i]] > B[winner]:
        winner = C[i]
```

### Task

For each  $i$  from 1 to  $N$ , determine in how many distinct ways Clin can give the requests such that the winning card is card  $i$ . Since the answer may be very large, output it modulo  $10^9 + 7$ .

### Input data

The first line contains the integer  $N$ . The second line contains  $N$  integers, the elements of permutation  $A$ . The third line contains  $N$  integers, the elements of permutation  $B$ .

### Output data

Print  $N$  integers, separated by spaces, the answer for each  $i$ .

## Constraints

- $1 \leq N \leq 200\,000$ .

| # | Points | Constraints                                              |
|---|--------|----------------------------------------------------------|
| 1 | 6      | $A = 1, 2, \dots, N$ and $B = N - 1, N - 2, \dots, 1, N$ |
| 2 | 6      | $1 \leq N \leq 10$                                       |
| 3 | 14     | $1 \leq N \leq 20$                                       |
| 4 | 13     | $1 \leq N \leq 500$                                      |
| 5 | 16     | $A$ and $B$ are generated uniformly at random            |
| 6 | 24     | $1 \leq N \leq 5000$                                     |
| 7 | 21     | No additional restrictions                               |

## Examples

| stdin               | stdout | Explanations                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3<br>2 1 3<br>3 2 1 | 4 0 2  | $C = [1, 2, 3] \Rightarrow$ At the end of the algorithm, winner = 1<br>$C = [1, 3, 2] \Rightarrow$ At the end of the algorithm, winner = 1<br>$C = [2, 1, 3] \Rightarrow$ At the end of the algorithm, winner = 1<br>$C = [2, 3, 1] \Rightarrow$ At the end of the algorithm, winner = 1<br>$C = [3, 1, 2] \Rightarrow$ At the end of the algorithm, winner = 3<br>$C = [3, 2, 1] \Rightarrow$ At the end of the algorithm, winner = 3 |